DASHFLOW

PRD 3.3 — Automated Planning & Order Dispatch Engine

Product Requirements Document · portfolio sample. Company, client, and product names have been genericized; structure and logic are unchanged.

- **Status:** Draft — re-baselined to the confirmed template model (user validation + BRD PP-01..03, 27 Jul); supersedes the registry model
- **Owner:** Jauhar Nurandyo
- **Epic:** Project Brief - EPIC 3 - Production Planning
- **Phase:** P1 · the definition layer — the **Planning Template** (BRD: *Mapping Kategori PP*) that 3.2's generation, 3.1's actual capture, and the Production Order all resolve against
- **Tenant scope:** the client — QSR chain, QSR-company whitelisted
- **Model:** exit-only deduction (EPIC 2) — stock deducts at **sale and waste** via BOM explosion; production banks stock onto the **produced output**. The template carries no stock and no quantities; it defines *what is planned together* and *in what time-shape*
- **Surface: Planning Template** menu (HO-side), in the Production Planning module; consumed by plan generation (3.2), the actual entry (3.1), and the Production Order

1. Objective

Define the one object every production plan is built from: the **Planning Template** — HO's mapping of what is produced together (BRD calls this the *Kategori PP*; this system says **template** throughout). A template groups, under one free name, the **output it produces** — an FG or a produced SFG — (one or several) and the **menus that consume that output** (with their BOM conversion), and fixes the template's **cycle** (Hourly or Daypart) and unit. Planning happens at the template level — one production number per template per slot; stock lands on the **produced output**; sales are counted through the **menu members**. This PRD holds no quantities and moves no stock — it is the definition layer everything else resolves against.

> **Scope boundary — what the template is and is not.** The template is a **planning grouping and a validation set**, not a deduction structure: what deducts at a sale or a waste is the **menu's / output's own BOM** (EPIC 2) — the template is never consulted at exit time. It hosts **references**, not masters: **End Shelf Life** is read from the product itself (PRD 1); **Waste Material** is a read-only tag into the Waste Management master (Accounting owns CRUD — HO views only). The generation math, buffer, and reference-date engine are 3.2; the actual entry and its output pick are 3.1; the **Production Order (one per plan, with per-slot tasks)** is 3.2/3.1's contract. BRD anchors: **PP-01, PP-02, PP-03** (create / edit / delete Mapping Kategori PP).

2. Problem Statements

- The same physical cook sells through many doors — Crispy Skin à la carte, as add-on, inside a combo — and planning each menu independently plans **one item twice**. The template exists to collapse those menus into **one production number** and count every sale against it.
- A money-and-menu world needs a bridge to a cook-and-stock world: sales happen in menu pcs, production happens in the output's units (grams, pcs). The mapping's per-menu **Qty BOM** is that conversion, and without it the reference-day autofill has nothing to compute with.

- Some templates produce **one output** (SKIN → Chicken Skin); some **several** (Fruit Tea → Pomelo, Blackcurrant, Jasmine Tea premixes). The template plans one total; the system must know **which output can receive** the production and how the total splits across them.
- The cycle (Hourly vs Daypart) shapes everything downstream — slots, units, batch math, the entry form — and it is a property of **how the template's output is cooked**, not of any single plan.

3. Goals

- One home for the **Planning Template**: free name, company, cycle (**Tipe**: Hourly | Daypart), UoM, station, **Template FG** (single or multi — an FG or a produced SFG), and **menu members** with their Qty BOM conversion — authored HO-side, per BRD PP-01..03.
- **Output-grain stock, template-grain planning:** the template plans one total; production banks onto a specific output under it; every menu member's sale deducts through its own BOM.
- **The split rule** for multi-output templates: **V1 splits evenly across eligible outputs**, with reference-mix weighting as a fast-follow; eligibility-gated, defined here as the contract 3.2's generation implements.
- **Structural constraint:** multi-output templates are **Daypart only**; an Hourly template carries exactly **one output**.
- **Editable definitions, forward-only:** template edits apply to planning from the current day onward (PP-02.5); submitted plans keep the definitions they were generated with.

4. The Model — one template, two sides

A template has a **production side** and a **demand side**, and the two must never be conflated:

Side	Field	Holds	Role
Production	**Template FG** (single or multi select)	The output this template produces — an FG or a produced SFG (Chicken Skin; the three Fruit Tea premixes; the BBQ Sauce levels)	What production banks stock onto; the valid options for 3.1's actual-entry output pick
Demand	**Members (Menu)**	The menus that consume this template's output, each with **Qty BOM** (menu pcs → output unit)	What the reference-day sales count through; what makes the template's demand one number

The aggregation direction is up, not down. Members exist so that every sale path of the same physical cook rolls into **one production total** — SKIN plans one Chicken Skin number whether it sold à la carte, as add-on, or in a combo. The template never plans menus.

Stock is output-grain. The template has no balance of its own — it has nothing to stick to (no master row, no UoM, no shelf life). Production banks PRODUCTION_STOCK_IN onto the **picked output**; a menu's sale deducts the output named in that menu's own BOM (EPIC 2). The template constrains the pick and groups the plan; it does not key stock and is not consulted at exit.

An FG and a produced SFG are the same kind of thing here. A produced SFG is a BOM-bearing cooked intermediate — the **BBQ Sauce levels** are the canonical case — batch-cooked ahead on **its own template**, with its own plan lines, tasks, actual cook, and stock-in, exactly like an FG. The FG-vs-SFG line is about **where it exits** (an FG deducts at its own sale; a produced SFG deducts when its parent FG sells or wastes), not how it is made. **The discriminator is BOM presence:** a procured SFG (no BOM, arrives via GR) is not produced and never appears as a template output.

Cycle is the template's (Tipe, per BRD PP-01.b). Daypart templates plan grams per window with a derived batch; Hourly templates plan pcs per clock-hour wave. A plan date simply contains whatever templates exist — Daypart and Hourly templates coexist on the same date; the plan has no cycle of its own.

Multi-output ⇒ Daypart only. A template with more than one output must be Daypart; Hourly templates carry exactly one output. (Hourly's per-wave, pcs-grain rhythm doesn't admit a within-wave split; Daypart's window totals do.)

The split rule (multi-output, Daypart) — the contract 3.2's generation implements:

Template total splits across its **eligible** outputs (eligible = has stock / not removed). **V1 splits evenly among eligible**; reference-mix weighting — cook more of what sold more — lands as a fast-follow once the per-output reference pipeline is proven. An ineligible output's share redistributes across the remaining eligible outputs in the same ratio. One eligible output takes the full total.

5. System Flow & Behavior

5.1 Create a template (HO) — BRD PP-01

"+ New Template" opens the mapping form:

Field	Required	Notes
Company	Yes	The brand/company the template belongs to — shown on the record
Nama Template	Yes	**Free text** — Fruit Tea, SKIN, Crispy Chicken WHOLE (CRISPY) — not picked from a list (BRD PP-01.c: *Input Kategori PP*)
Tipe	Yes	**Hourly** or **Daypart** — the template's cycle; validation: multi-output requires Daypart
UoM	Yes	The template's production unit (gr, pcs)
Station	Yes	The station this template's output cooks at — **mandatory**, single-select from the station master (**KDS Routing config, TBC**); consumed by the Production Order/KDS routing
Waste Material	Yes	Single-select, **read-only reference** into the Waste Management master (Accounting owns CRUD; HO views)
End Shelf Life	display	**Referenced from the product** (PRD 1) — shown for the output, not authored here
Template FG	Yes 1+	Single **or** multi select of FG **or produced-SFG** product_variant — what this template produces. Multi ⇒ Tipe must be Daypart
Members (Menu)	Yes 1+	The menus consuming this template, each with **Qty BOM** (auto-filled from the BOM: menu pcs → output unit)

Simpan publishes the template; created/modified stamps land in the change log (PP-01.3).

5.2 Edit & delete — forward-only (PP-02, PP-03)

Every field is editable; menu members and outputs can be added or removed. **Changes apply from the current planning day onward** (PP-02.5) — plans already generated/submitted keep the definitions they were built with. Delete warns if the template is in an open plan; it never touches submitted plans. All changes stamped.

Removing an output mid-day is the live case the split rule covers: the output becomes ineligible and its share redistributes across the remaining outputs at the next generation — down to one output taking the full total.

5.3 What the plannable gate is now

A template is plannable when it has **≥1 produced output and ≥1 member whose Qty BOM resolves**. No separate attributes registry — the registry model is retired (§9). No complete template → nothing to plan; a menu missing its BOM conversion is flagged at the template, not at plan time.

5.4 What consumes the template

- **3.2 generation:** reference-day sales are counted **through the members** (menu pcs × Qty BOM → template units), producing the template's slotted totals; multi-output totals split per §4's rule; the Raw Material preview groups **at output level per template**, never per menu.
- **3.1 actual entry:** the template is picked on the entry (form label: *Pilih Template*, renamed from the BRD's *Pilih Kategori PP*); the **output pick list = this template's outputs** (a single-output template resolves implicitly); the banked stock-in lands on that output.
- **Production Order / KDS:** orders carry the template's slotted lines; station routing reads the template's station.
- **EPIC 4:** expiry uses the output's product shelf life (referenced here for display); expiry/fixed-waste posts against the template's Waste Material tag; the WASTE exit explodes the output's BOM (EPIC 2). All SFG expire by EOD via WASTE — a produced SFG's stock does not carry overnight.
- **WST-15 anomaly monitoring:** uses this mapping to relate PP Actual and sales to a waste material — the template is the join key.

6. Data Model (replaces the registry)

Entity / Field	Notes
planning_template	One per company × template: nama (free text), company_id, tipe (HOURLY / DAYPART), uom, station_id (**required**, FK → station master / KDS Routing config, TBC), waste_material_id (FK → Waste master, read-only here)
planning_template_fg	One per template × output: template_id × product_variant_id (**FG or produced SFG** — name retained for continuity). **Constraint: count > 1 ⇒ template.tipe = DAYPART**
planning_template_member	One per template × menu: template_id × menu variant, qty_bom (menu pcs → template UoM; sourced from the BOM)
Shelf life	**Not stored** — read from the output's product master (PRD 1)
plannable_item_attributes	**Retired** — dropped with the registry model (§9)
created/updated at/by	On all entities; every change stamped (PP-01..03)

7. User Stories & Acceptance Criteria

Story 1 — HO maps a single-output template

As HO, I map SKIN once — its output and the menus that sell it — so its every sale path plans as one production number.

Acceptance Criteria

- The form takes company, free-text name, Tipe, UoM, station, waste-material reference, **one output** (Chicken Skin), and menu members each with Qty BOM auto-filled from the BOM.
- The output's shelf life displays from the product master; it is not editable here.
- A produced SFG is authored the same way — BBQ Sauce Lv 4 is selected as the output; a procured SFG (no BOM) is never offered as an output.
- Simpan publishes the template; created/modified stamps land in the change log (PP-01.3).

```
Scenario: A produced SFG is a valid template output
  Given HO maps BBQ SAUCE LV4 with output "BBQ Sauce Lv 4" (a BOM-bearing SFG)
  When HO saves the template
  Then the template is accepted and behaves as any FG template
  And a procured SFG with no BOM is not offered as a template output
```

Story 2 — Demand resolves through the members, not the menus

As HO, I attach every menu that sells the output as a member, so all its sale paths roll into one production total and no menu plans on its own.

Acceptance Criteria

- Reference-day demand for a template = Σ (each member's sold pcs × its Qty BOM) — one number.
- No member menu ever appears as its own plan line; aggregation is **up** (members → one total), never down.
- Members span à la carte, add-on, and combo lines; each carries its own Qty BOM.

```
Scenario: Three sale paths, one plan line
  Given SKIN maps output Chicken Skin with members "Crispy Skin", "Add-on Crispy
Skin", "Crispy Skin in Combo"
  When the plan for Friday generates
  Then SKIN appears as one template with one slotted total
  And no member menu appears as its own plan line
```

Story 3 — HO maps a multi-output template (Daypart only)

As HO, I map Fruit Tea with its three premix outputs, so one template total covers whichever flavours the outlet actually makes.

Acceptance Criteria

- Template FG accepts multiple outputs; selecting >1 output **requires Tipe = Daypart** — choosing Hourly with multiple outputs is blocked with a message.
- Removing an output applies forward-only (PP-02.5) and makes it ineligible from the current planning day; submitted plans keep their snapshot.
- How the template total divides across its outputs is specified in Story 4.

```
Scenario: Multi-output forced to Daypart
  Given HO selects Pomelo, Blackcurrant, and Jasmine Tea as Template FG
  When HO sets Tipe = Hourly
  Then save is blocked: "Multi-output templates run on Daypart"
```

Story 4 — The multi-output total splits across eligible outputs

As an SM, when a multi-output template plans a total, the system splits it across the outputs I can actually make, so the plan reflects available stock without me apportioning by hand.

Acceptance Criteria

- The template total splits across its **eligible** outputs (eligible = has stock / not removed), per §4's split rule.
- **V1 splits evenly** among eligible outputs; reference-mix weighting is a **fast-follow** once the per-output reference pipeline is proven.
- An ineligible output's share redistributes across the remaining eligible outputs in the same ratio; one eligible output takes the full total.
- The split is frozen at generation and recomputes only when eligibility changes.

```
Scenario: V1 even split, and redistribution on removal
  Given Fruit Tea totals 90 with three eligible outputs
  Then generation allocates 30 / 30 / 30
  When Jasmine Tea is removed (or has no stock)
  Then the 90 reallocates Pomelo 45 / Blackcurrant 45

Scenario: Reference-mix weighting (fast-follow, when the per-output pipeline is live)
  Given Fruit Tea's reference mix is Pomelo 45 / Blackcurrant 40 / Jasmine Tea 15 and
the total is 100
  Then generation allocates 45 / 40 / 15
```

Story 5 — The gate blocks an incomplete template

As an SM, I can only plan templates whose mapping is complete, so no plan produces output the system can't convert, bank, or waste-route.

Acceptance Criteria

- A template is offered to planning only when it has ≥1 produced output and ≥1 member with a resolvable Qty BOM.
- A member whose BOM cannot resolve flags the **template** with what's missing; the template is excluded from generation until fixed.
- Template edits and deletes are forward-only (PP-02.5, PP-03.4); submitted plans keep their snapshot.

```
Scenario: Missing conversion blocks the template, names the menu
  Given Fruit Tea's member "Blackcurrant 1L" has no Qty BOM
  When the plan generates
  Then Fruit Tea is excluded from generation and its template shows "Blackcurrant 1L:
BOM conversion missing"
```

8. Dependencies

Dependency	Direction	Note
EPIC 2 BOM + exit engine	upstream	Qty BOM conversions; deduction truth at sale/waste — the template is never consulted at exit
PRD 1 product master	upstream	FG and produced-SFG variants exist with **shelf life on the product**; menus exist as sale variants
Station master (KDS Routing config)	upstream (**TBC**)	Mandatory station field sources from it; master ownership/shape to be confirmed — named cross-PM dependency
Waste Management master (EPIC 4 / Accounting)	upstream	Waste Material tag target — **Accounting owns CRUD, HO read-only** (deviation of ownership vs BRD's Ops-HO framing noted for Kevina)

Dependency	Direction	Note
PRD 3.2 generation	downstream	Counts demand through members; implements the split rule; groups the material preview at output-per-template
PRD 3.1 actual entry	downstream	Output pick list = the template's outputs; banked stock lands on the picked output
Production Order / KDS	downstream	**One Production Order per plan**; its **tasks** are per slot and carry the template's slotted lines; station routing reads the template's station
WST-15 monitoring	downstream	Uses the mapping as the PP-Actual ↔ sales ↔ waste join key

9. Risks & Assumptions

Item	Type	Severity	Note
Template mapping (BRD: Kategori PP) restored as the spine	**Decided (user validation + BRD, 27 Jul)**	—	Reverses the 16 Jul "Kategori PP replaced by Template + registry" re-baseline. The registry (plannable_item_attributes) is retired: shelf life lives on the product (PRD 1), waste material is a read-only reference to the Waste master, station moves onto the template. The 16 Jul detour and its reversal both travel to Kevina as context. **On push, the superseded rev-32 model summary is attached as an Outline comment on this doc for change tracking**
Produced SFG is a valid template output	**Decided (PM, 28 Jul)**	—	A BOM-bearing cooked intermediate (BBQ Sauce levels) sits on its own template with its own plan lines, tasks, actual cook and stock-in — identical to an FG. The FG/SFG line is about **where it exits**, not how it is made. **Discriminator = BOM presence:** a procured SFG (no BOM, arrives via GR) is never a template output. Folded into §4/§5/§6 rather than carried as a separate section
Produced SFG banks to its own ledger and expires daily	**Decided (PM, 28 Jul)**	—	The only two behavioural differences from an FG: an SFG stock-in lands on the SFG ledger, and **all SFG expire by EOD via WASTE** (explode + deduct) — SFG stock never carries overnight
Modifier-borne SFG must live in the parent FG's BOM	Assumption / data discipline	Medium	e.g. Crispy Chicken + "Lv 4" → the FG's BOM includes BBQ Sauce Lv 4, so a sale walks FG → SFG Lv 4 → RM. A sauce level sold on its own would be an FG instead. Authoring requirement flagged to **Kevina / Josafat**
Output-grain stock; the template never holds a balance	**Decided (PM, 27 Jul)**	—	Production banks onto the picked output; sale/waste deduct the output named in the menu's/output's own BOM. Supersedes the interim category-pool reading. Cost accepted: none — this matches how every epic keys stock

Item	Type	Severity	Note
Multi-output ⇒ Daypart only	**Decided (PM, 27 Jul)**	—	Hourly templates carry exactly one output; enforced at the template form. Simplifies 3.1's hourly entry (no output pick on hourly)
Split rule: V1 even split over eligible; mix-weighting fast-follow	**Decided (PM, 27–28 Jul)**	Medium	V1 ships even split (total ÷ eligible) — app-side, no data-team dependency. Reference-mix weighting lands once the per-output reference pipeline is proven (data-team owned). The split is frozen at generation (3.2), recomputed only if the acuan date changes
Menu→output link must be unambiguous in every member's BOM	Assumption / data discipline	Medium	A member menu must name exactly which output it draws (Blackcurrant 16OZ → Blackcurrant premix); ambiguity breaks output-grain deduction. Authoring requirement flagged to Kevina/ops
Per-item availability within a multi-output template	Accepted trade	Low	The plan manages the template total; whether the outlet made enough *Pomelo specifically* is visible only through output stock, not the plan. Matches the BA's "regardless apa yg masuk"
Station master (KDS Routing config) is TBC	Assumption / cross-PM	Medium	The field is mandatory now but its master doesn't exist yet — template creation is gated on at least a provisional station list; confirm owner and shape
"Template" replaces "Kategori/Category" as the running term	**Decided (PM, 27 Jul)**	—	"Category" collides with menu categories, product categories, and procurement Category; template is unclaimed and more accurate (the object prescribes, not classifies). BRD alias stated once (*Mapping Kategori PP*); user-facing labels renamed incl. 3.1's *Pilih Template*. Label deviation carried to Kevina
Waste Material ownership	Deviation (feedback #14)	Low	BRD frames the master as Ops-HO; users ruled Accounting owns CRUD, HO views. Carried to Kevina